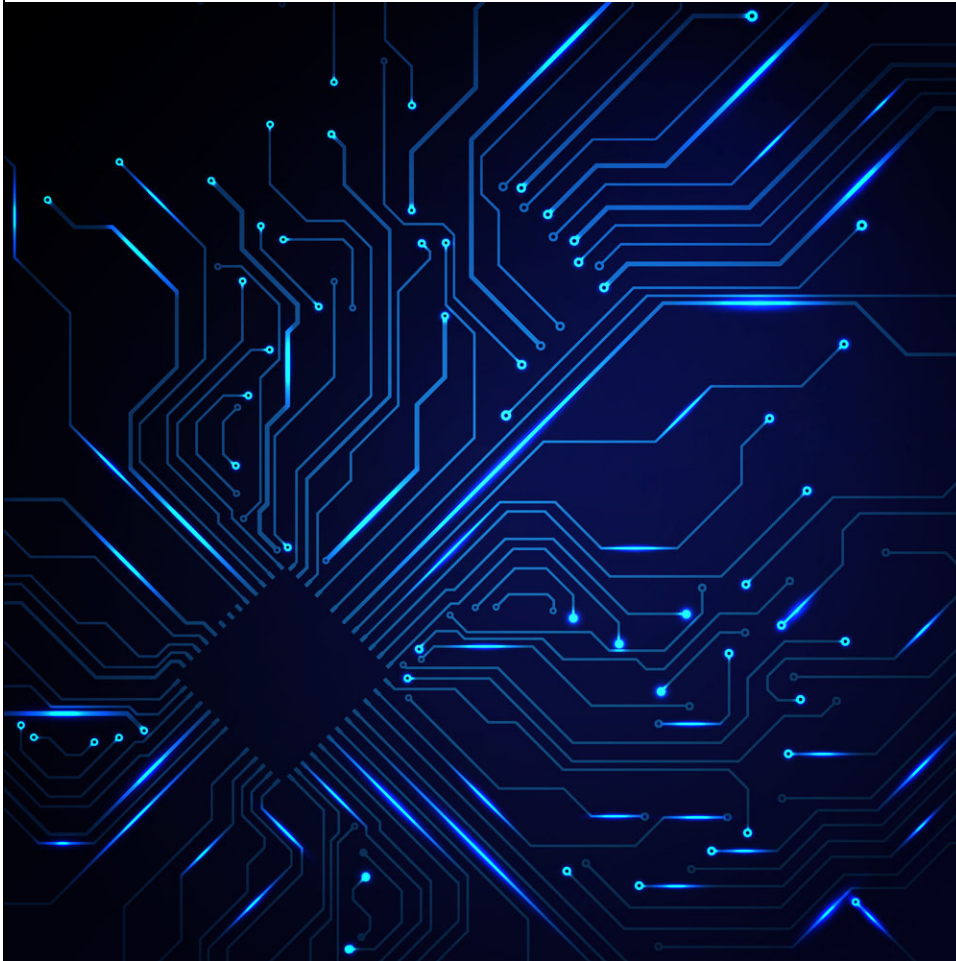




LAB 1

BASIC LOGIC GATE IMPLEMENTATION

Group : 14

Name : 黃翊宏

Student ID : 01357101

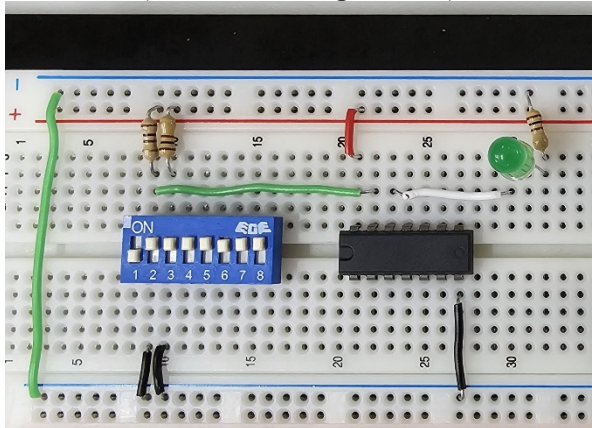
Notice

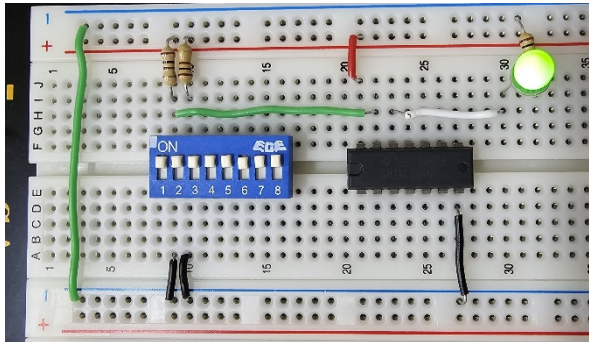
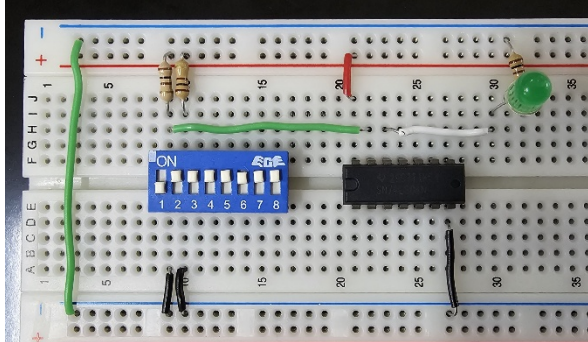
1. Lab report must be submitted in PDF format.
2. Lab report must be submitted to the TronClass by 13:00 next class hour.
3. Each student must submit one lab report.
4. File name of the lab report must be in Lab#_StudentID.

Lab 1-1:

- 7404 (NOT gate) : $F = A'$

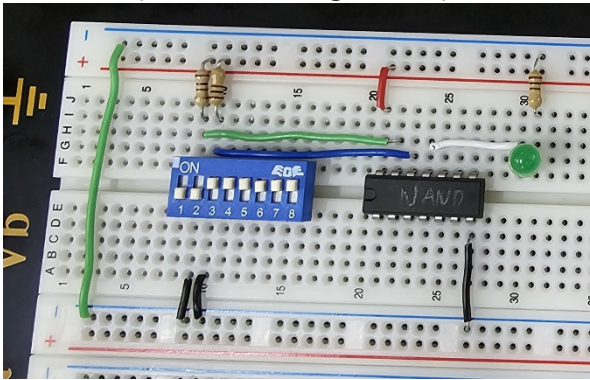
Truth Table (Measured Voltage) Output with LED Connected		Truth Table (Measured Voltage) Output without LED Connected	
A	F	A	F
0	2.96	0	1.89
1	0.15	1	0.01

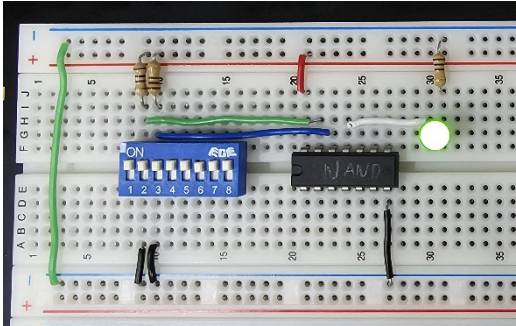
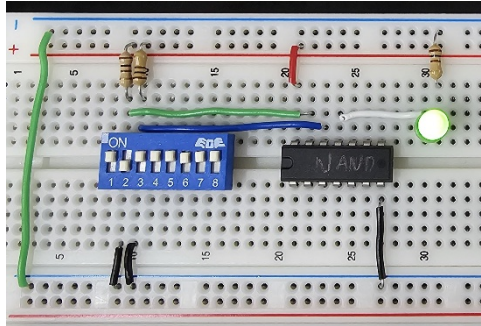
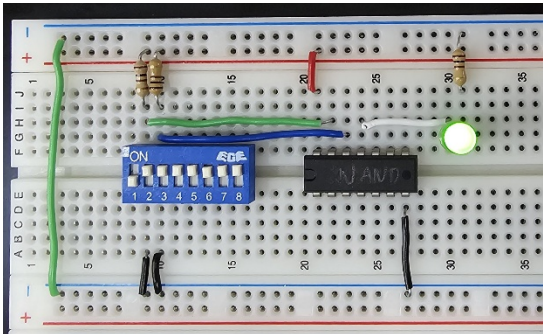
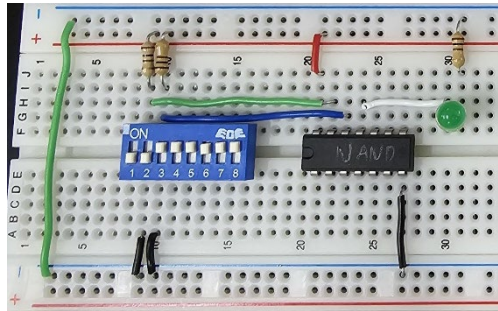
Photo of the Wiring of the Breadboard (With LED, Unpowered)		Truth Table (Logic)	
		A	
0		1	
1		0	

Results with LED, Powered			
A = 0		A = 1	
Photo of the Result		Photo of the Result	
			

- 7400 (NAND gate) : $F = (A \cdot B)'$

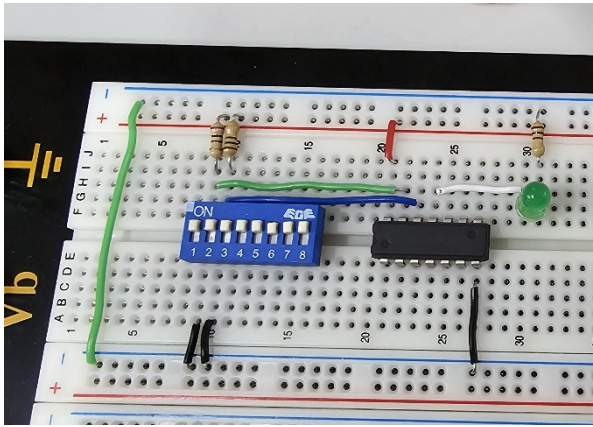
Truth Table (Measured Voltage) Output with LED Connected			Truth Table (Measured Voltage) Output without LED Connected		
A	B	F	A	B	F
0	0	2.90	0	0	1.75
0	1	2.91	0	1	1.76
1	0	2.90	1	0	1.76
1	1	0.12	1	1	0.18

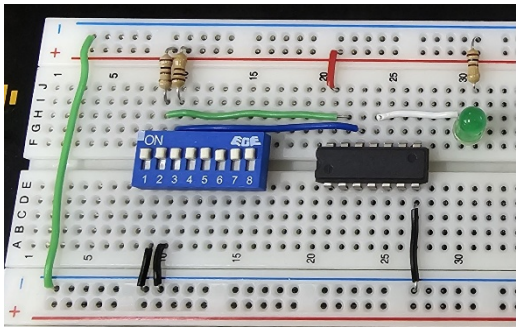
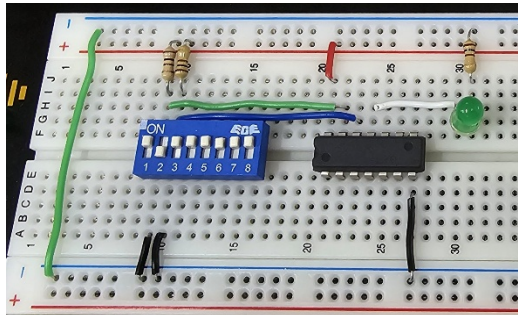
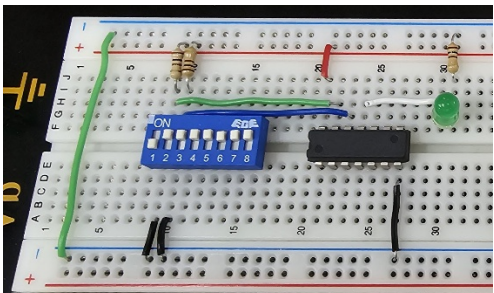
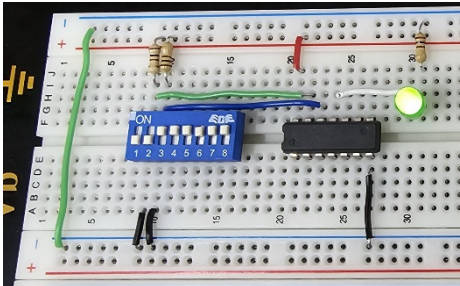
Photo of the Wiring of the Breadboard (With LED, Unpowered)			Truth Table (Logic)		
			A	B	F
			0	0	1
			0	1	1
			1	0	1
			1	1	0

Results with LED, Powered					
A = 0, B = 0			A = 0, B = 1		
Photo of the Result			Photo of the Result		
					
A = 1, B = 0			A = 1, B = 1		
Photo of the Result			Photo of the Result		
					

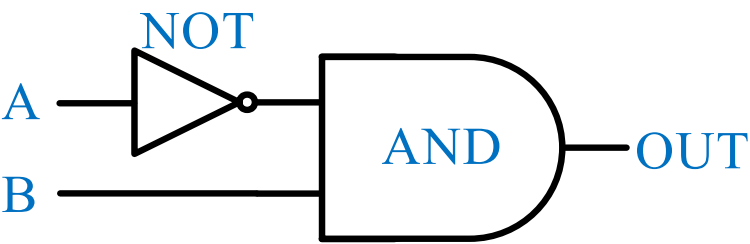
- 7408 (AND gate) : $F = A \cdot B$

Truth Table (Measured Voltage) Output with LED Connected			Truth Table (Measured Voltage) Output without LED Connected		
A	B	F	A	B	F
0	0	0.15	0	0	0.07
0	1	0.15	0	1	0.07
1	0	0.15	1	0	0.07
1	1	2.99	1	1	2.02

Photo of the Wiring of the Breadboard (With LED, Unpowered)			Truth Table (Logic)		
			A	B	F
			0	0	0
			0	1	0
			1	0	0
			1	1	1

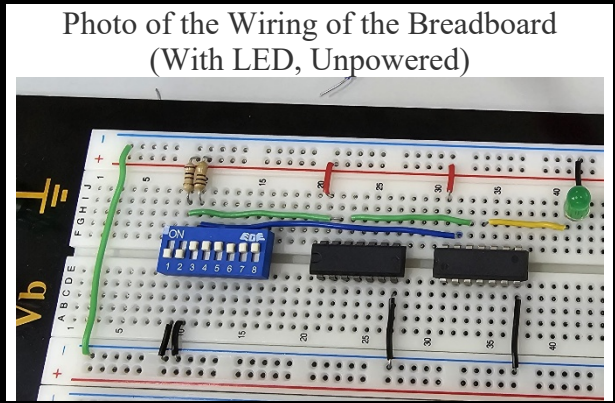
Results with LED, Powered					
A = 0, B = 0			A = 0, B = 1		
Photo of the Result 			Photo of the Result 		
A = 1, B = 0			A = 1, B = 1		
Photo of the Result 			Photo of the Result 		

Lab 1-2:



Truth Table (Measured Voltage) Output with LED Connected		
A	B	OUT
0	0	0.15
0	1	2.24
1	0	0.15
1	1	0.15

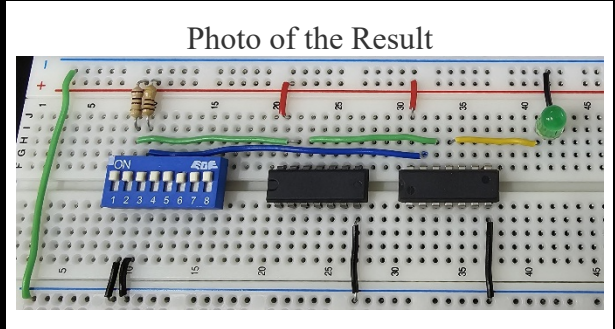
Truth Table (Measured Voltage) Output without LED Connected		
A	B	OUT
0	0	0.05
0	1	1.79
1	0	0.05
1	1	0.05



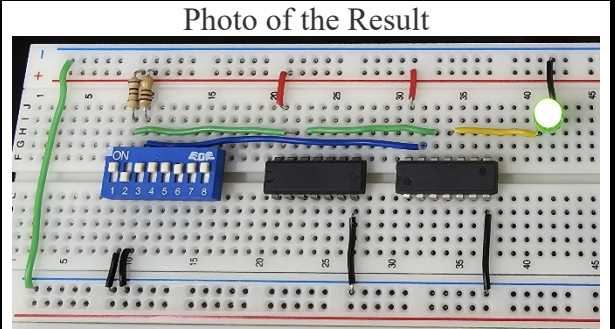
Truth Table (Logic)		
A	B	OUT
0	0	0
0	1	1
1	0	0
1	1	0

Results with LED, Powered

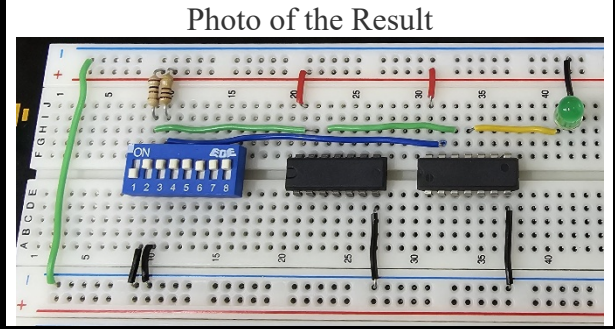
A = 0, B = 0



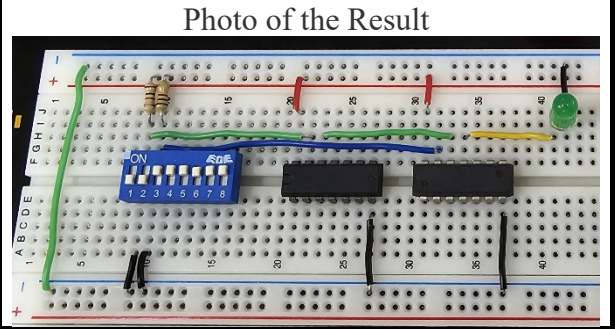
A = 0, B = 1



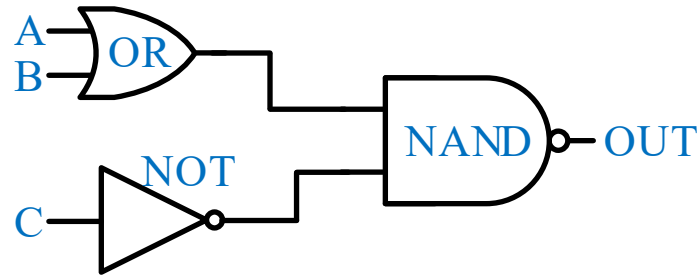
A = 1, B = 0



A = 1, B = 1

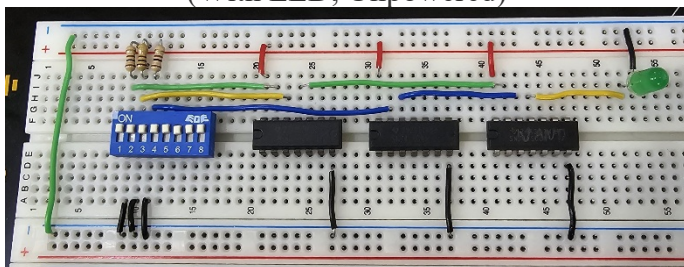


Lab 1-3:



Truth Table (Measured Voltage) Output with LED Connected				Truth Table (Measured Voltage) Output without LED Connected			
A	B	C	OUT	A	B	C	OUT
0	0	0	2.18	0	0	0	1.75
0	0	1	2.18	0	0	1	1.76
0	1	0	0.13	0	1	0	0.05
0	1	1	2.19	0	1	1	1.76
1	0	0	0.13	1	0	0	0.06
1	0	1	2.19	1	0	1	1.76
1	1	0	0.13	1	1	0	0.05
1	1	1	2.19	1	1	1	1.76

Photo of the Wiring of the Breadboard
(With LED, Unpowered)



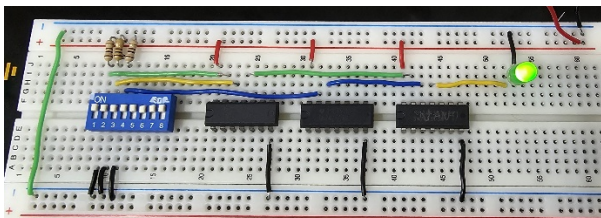
Truth Table (Logic)

A	B	C	OUT
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Results with LED, Powered

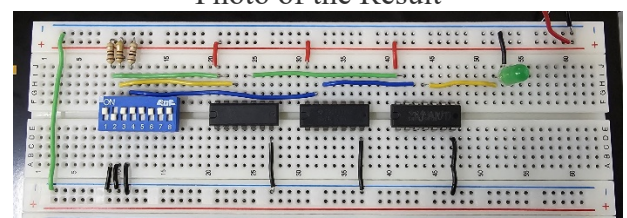
A = 0, B = 0, C = 0

Photo of the Result



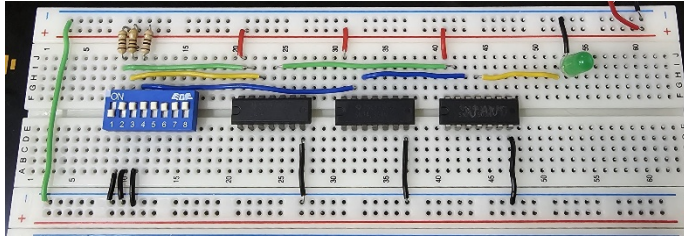
A = 0, B = 1, C = 0

Photo of the Result



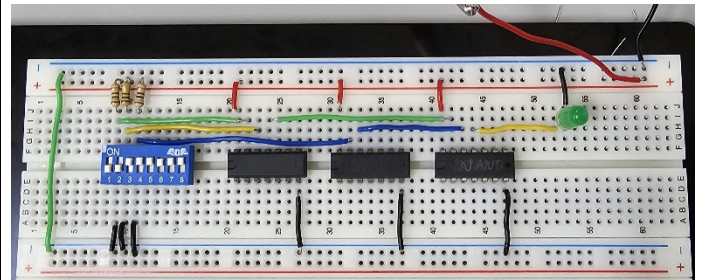
$A = 1, B = 0, C = 0$

Photo of the Result



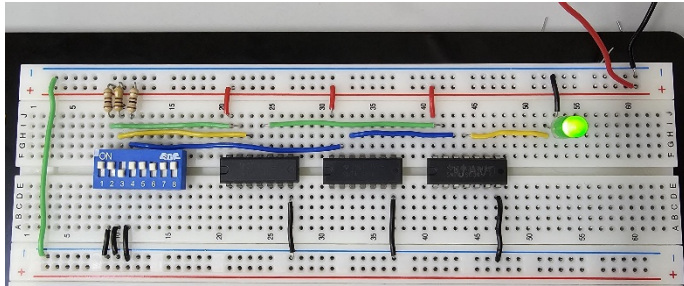
$A = 1, B = 1, C = 0$

Photo of the Result



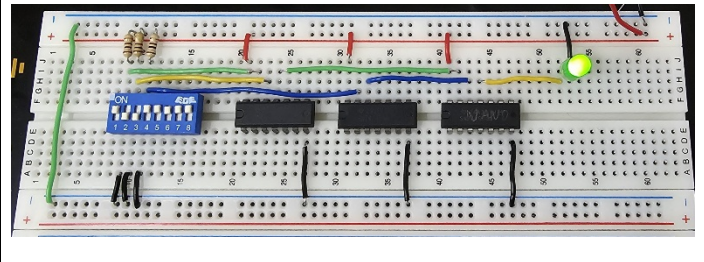
$A = 0, B = 0, C = 1$

Photo of the Result



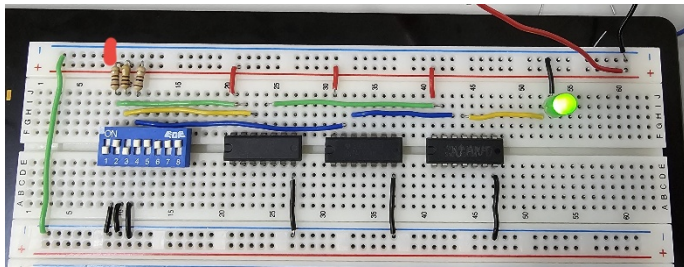
$A = 0, B = 1, C = 1$

Photo of the Result



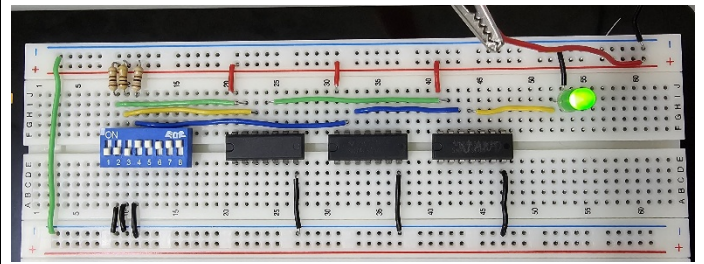
$A = 1, B = 0, C = 1$

Photo of the Result



$A = 1, B = 1, C = 1$

Photo of the Result



● Lab Summary and Reflection :

Lab Summary:

In this laboratory experiment, we engaged in the practical implementation and testing of basic digital logic circuits using TTL integrated circuits—specifically, the 7404 (NOT gate), 7400 (NAND gate), and 7408 (AND gate). The experiment was structured into three segments: Lab 1-1 focused on constructing and evaluating individual logic gates, Lab 1-2 involved assembling a composite logic circuit, and Lab 1-3 integrated multiple inputs to implement more complex logical operations. For each segment, circuits were assembled on breadboards, and detailed measurements were taken for the output voltages both with and without LED connections. These measurements were then compared against the theoretical truth tables to verify the proper functionality of the circuits. In addition, photographic documentation of the wiring setups and the resulting outputs was captured to provide a visual confirmation of the experimental procedures and outcomes.

In class, the professor introduced the physical structure of the TTL device in detail, and voltage logic levels of TTL devices. This information was crucial in guiding the practical aspects of circuit assembly and troubleshooting. Overall, the experiment successfully bridged the gap between theoretical concepts and hands-on application, reinforcing core principles of digital logic design and circuit analysis.

Reflection:

The laboratory experience provided several valuable insights, both technically and logistically. One of the primary challenges was the insufficient availability of IC chips—many of which were either damaged or missing. This shortage forced several groups to borrow components from one another, resulting in delays and added complexity to the experimental process. Furthermore, the limited number of teaching assistants contributed to slower resolution of queries and troubleshooting issues during critical stages of the experiment, further hindering progress.

In response to challenges encountered in the previous week(Lab0) —such as the issues with faulty breadboards—this experiment incorporated significant procedural improvements. The use of personal breadboards ensured a more reliable and consistent assembly process, thereby minimizing the impact of equipment failures. Additionally, enhanced coordination among team members through a more structured division of labor and improved progress planning proved instrumental in streamlining the experiment. This experience underscored the importance of effective resource management, proactive troubleshooting, and robust teamwork in a laboratory setting. Overall, the lab not only deepened my technical understanding of digital logic but also highlighted the critical role of adaptability and communication in overcoming practical challenges.